

HARSHA WADUTHANTHRI

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As a passionate, enthusiastic, and dedicated Data Science graduate awaiting final results, I have a keen interest in data analysis, machine learning, and data engineering. Having completed all requirements for a Bachelor's degree and currently awaiting final results, I have acquired a solid foundation in data science principles and techniques. Eager to apply my knowledge to real-world projects, I am actively seeking job opportunities to further develop my skills in data science. Committed to leveraging innovative data-driven solutions to solve complex problems, I continuously seek opportunities to expand my expertise and connect with like-minded professionals in the field.

EDUCATION

BSc (Hons) in Information Technology – Specialization in Data Science (Awaiting Final Results)
Sri Lanka Institute of Information Technology (SLIIT), Malabe 2022 – Present

Higher Diploma in Information Technology
Sri Lanka Institute of Information Technology **Completed** : July 2024

Advanced Level 2021 – Physical Science Stream
St.Sylvester's College, Kandy 2007-2021

EXPERIENCE

AI and Backend development Intern 2025 Sep – 2026 Mar
SparQ Corporate

- Developed and maintained microservice-based backend services for authentication, inventory management, onboarding workflows, and multi-tenant schema handling using Spring Boot and PostgreSQL.
- Contributed to AI-powered backend features by integrating Google Gemini API, implementing AI-assisted workflows, and exposing AI functionalities through REST APIs.
- Built secure and scalable backend systems with optimized business logic and API handling.
- Tools and Technologies – Spring Boot, Java, PostgreSQL, REST APIs, Prompt Engineering, Microservices, GitHub, Postman, Google Gemini API, Ollama, AWS

PROJECTS

Data Warehousing Project 2025 - May

- Designed and implemented an end-to-end BI solution with ETL, data modeling, and reporting.
- Built SSIS ETL workflows with SCD Type 2 for historical data tracking.
- Modeled a Snowflake Schema (1 fact, 5 dimensions) for scalable analytics.
- Developed SSAS OLAP cube with calculated measures and hierarchies.
- Integrated SSAS with Power BI to build interactive dashboards.
- Tools and Technologies - SQL Server, SSIS, SSAS, Power BI, Visual Studio, Excel

IoT-Based Predictive Maintenance System for Industrial Sewing Machines 2026

- Developed an IoT-based predictive maintenance system using multi-sensor data monitoring.
- Implemented real-time RPM monitoring and MQTT-based data transmission with offline buffering support.
- Built machine learning models for anomaly detection and machine degradation analysis.
- Developed dashboards for monitoring machine health, vibration, temperature, and production metrics.
- Tools and Technologies - ESP32, MQTT, MongoDB, Python, scikit-learn, React.js, Node.js, WebSocket

SewMetrics Analytics Dashboard with AI Assistant	2026
• Developed an analytics dashboard for textile manufacturing data visualization and machine monitoring. • Integrated an AI-powered chatbot for natural language queries and machine troubleshooting support. • Implemented real-time analytics and data visualization using FastAPI and chart components. • Tools and Technologies – React.js, FastAPI, Pandas, PyTorch, HuggingFace Transformers, Recharts, Python	

Video Game Sales Prediction Model (Deployed Web App Link)	2024
• Developed a Streamlit-based web app to predict global video game sales using a trained Random Forest model. • Performed extensive preprocessing and feature engineering. • Deployed via GitHub for real-time, user-friendly predictions. • Tools and Technologies - Jupyter Notebook, Streamlit, GitHub, Python	

SKILLS SUMMARY

- **Programming Languages:** Python, Java, C, C++, R, Kotlin
- **Databases:** MySQL, SQL, MongoDB, Oracle (ORDBMS - Learning)
- **Machine Learning:** Supervised Learning, Unsupervised Learning, Natural Language Processing (NLP)
- **Cloud Platforms:** AWS, Microsoft Azure
- **Libraries:** Pandas, NumPy, Matplotlib, Sci-kit Learn, PyTorch, TensorFlow, Seaborn
- **Visualization:** MS Excel, MS Power BI
- **Framework Tools:** Google Colab, Jupyter, Streamlit, Hadoop, Git
- **Soft Skills:** Critical Thinking, Analytical and Problem-Solving, Adaptability to new technologies, Teamwork, Communication, Planning, Innovation

TECHNICAL

- | | |
|---|---|
| • Machine Learning & Optimization | • Data Warehouse & Business Intelligence |
| • Visual Analytics and User Experience Design | • Internet of Things and Big Data Analytics |
| • Cloud Computing | • Data Structures & Algorithms |
| • Fundamentals of Data Mining | • Probability & Statistics |
| • Information Retrieval & Web Analytics | • Database Management System |

CERTIFICATES

• AWS Educate Machine Learning Foundations – AWS	2025
• SQL (Basic) Certificate – HackerRank	2025
• Power BI Beginner to Pro Workshop – Pragmatic Works	2025
• Python For Beginners – University of Moratuwa	2024
• AI/ML Engineer - Stage 1 – SLIIT	2024
• Power BI for Beginners – Simplilearn	2024

REFERENCES

K.R.C Koswatte

Manager, Academic Affairs/Lecturer
 EWIS Career Training (Pvt) Ltd, SLIIT
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